

ADARSH KUMAR SINGH

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Results-driven Full Stack AI Software Engineer specializing in Large Language Models (LLMs), Agentic Workflows, and scalable Python architectures. Proven ability to engineer autonomous multi-agent systems, implement Retrieval-Augmented Generation (RAG) pipelines, and build robust enterprise LLM guardrails from the ground up. Seeking an AI Engineering role to leverage expertise in Python, FastAPI, and GenAI to build secure, highly automated applications.

SKILLS

Programming Languages: Python, JavaScript, SQL, Bash

AI & Machine Learning: Large Language Models (LLaMA 3, GPT), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration (ReAct Loops), AI Guardrails, Prompt Engineering

Frameworks & Libraries: FastAPI, Pytest, Sentence-Transformers, ChromaDB (Vector Databases), GSAP, Three.js

Cloud & Infrastructure: Oracle Cloud Infrastructure (OCI), Oracle Database Design, Data Warehousing, Server-Sent Events (SSE)

Developer Tools: Git, REST APIs, JSON Validation, CI/CD Pipelines

ENGINEERING EXPERIENCE

Full Stack AI Engineer (Independent Capstone)

AI-Apex-Paragon Project | Jan 2024 - Present

- Engineered an autonomous multi-agent orchestration framework (ReAct Loops) from scratch to intercept CI/CD pipeline failures and automatically remediate broken codebase branches, reducing manual debugging time by 90%.
- Implemented a continuous diagnostic loop utilizing LLaMA 3 to analyze raw error logs, inject Python patches via custom file operations, and verify fixes using local sandboxed testing environments.
- Designed dynamic agent personas (Diagnoser, Developer, Verifier, PR Reviewer) to enforce zero-touch bug remediation across 100% of tested workflows, seamlessly integrating with the GitHub API.

Backend AI Developer (Independent Capstone)

AI-Apex-Paragon Project | Nov 2023 - Dec 2023

- Developed a highly secure compliance AI agent equipped with persistent state memory and rigorous LLM security boundaries, ensuring safe deployment in enterprise scenarios.
- Designed pre-execution Input Guardrails to prevent adversarial prompt injection and post-execution Output Guardrails to guarantee strict domain-scope adherence for all LLM responses.
- Built an automated LLM-as-a-Judge Evaluation Dashboard featuring Dynamic Model Failover routing, guaranteeing 99.9% uptime during strict API rate-limit bottlenecks.

AI Software Engineer (Independent Capstone)

AI-Apex-Paragon Project | Sep 2023 - Oct 2023

- Architected a Retrieval-Augmented Generation (RAG) service capable of ingesting enterprise PDFs and generating cited, accurate answers to eliminate LLM hallucination.
- Exposed the RAG service via a FastAPI backend, utilizing robust semantic search and semantic chunking strategies (Sentence-Transformers, ChromaDB) to retrieve the highest-context data dynamically.

Software Engineer (Independent Capstone)

AI-Apex-Paragon Project | Jul 2023 - Aug 2023

- Built a microservice that connects directly to the GitHub API, fetches raw pull request diffs, and streams changes to a high-reasoning LLM for deep architectural and code review.
- Enforced strict JSON schema generation to programmatically flag bugs, security vulnerabilities, and logic flaws that standard static analysis tools fail to catch.

EDUCATION

Dr. Abdul Kalam Technical University

B. Tech in Electronics and Communication Engineering | Aug 2022 - May 2026

CERTIFICATIONS

- Oracle GenAI Certified Professional (2025)
- Oracle Data Science Professional Certification (2025)
- IoT and Industrial Automation Certification (2025)